

Python Interview Preparation Guide

Fresher Edition by Aniket Verma

This comprehensive guide covers 11 sections of Python Interview preparation with detailed explanations, examples, and outputs in simple beginner-friendly language.

SECTION 1 – Core Python Basics

What is Python?

Answer: Python is a high-level, interpreted programming language. It is easy to read and write, and supports multiple programming styles like procedural and object-oriented programming.

Example code: See coding examples in PDF

Output: [See example output](#)

What are the key features of Python?

Answer: Python is simple, open-source, portable, dynamically typed, and has a large standard library.

Example code: See coding examples in PDF

Output: [See example output](#)

What is PEP 8?

Answer: PEP 8 is the official Python style guide which explains how to write clean and readable Python code.

Example code: See coding examples in PDF

Output: [See example output](#)

Difference between mutable and immutable data types?

Answer: Mutable objects like lists and dictionaries can be changed after creation. Immutable objects like strings and tuples cannot be changed after creation.

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 2 – Strings and Data Structures

How to reverse a string?

Answer: You can reverse a string using slicing.

Example code: See coding examples in PDF

Output: [See example output](#)

Difference between list, tuple, and set?

Answer: Lists are mutable and ordered, tuples are immutable and ordered, sets are mutable and unordered with no duplicates.

Example code: See coding examples in PDF

Output: [See example output](#)

How to remove duplicates from a list?

Answer: You can use set() to remove duplicates or list comprehension with condition checking.

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 3 – Functions and Scope

What is a function?

Answer: A function is a block of code that performs a specific task and can be reused.

Example code: See coding examples in PDF

Output: [See example output](#)

What are *args and **kwargs?

Answer: *args allows passing multiple positional arguments. **kwargs allows passing multiple keyword arguments.

Example code: See coding examples in PDF

Output: [See example output](#)

What is recursion?

Answer: Recursion is when a function calls itself to solve a smaller part of the problem until a base condition is met.

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 4 – Object-Oriented Programming (OOP)

What is a class and object?

Answer: A class is a blueprint for objects. An object is an instance of a class.

Example code: See coding examples in PDF

Output: See example output

What is `__init__`?

Answer: `__init__` is a constructor method in Python that initializes an object's attributes when it is created.

Example code: See coding examples in PDF

Output: See example output

Explain inheritance.

Answer: Inheritance allows a class to inherit attributes and methods from another class, promoting code reuse.

Example code: See coding examples in PDF

Output: See example output

SECTION 5 – File Handling & Exception Handling

How to open a file?

Answer: You can use `open('filename', 'mode')` to open a file. Modes can be 'r' for reading, 'w' for writing, 'a' for appending.

Example code: See coding examples in PDF

Output: [See example output](#)

How to handle exceptions?

Answer: Use try-except blocks. Example: `try: code except ExceptionType: handle error`

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 6 – Iterators, Generators, and Modules

What is a generator?

Answer: A generator is a function that returns an iterator using 'yield'. It generates values one by one instead of storing the whole sequence in memory.

Example code: See coding examples in PDF

Output: [See example output](#)

What is the difference between iterator and iterable?

Answer: An iterable can return an iterator, an iterator is an object with `__next__()` method that returns items one by one.

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 7 – Advanced / Common Coding Problems

Reverse a string 'Python'?

Answer: You can reverse using slicing.

Example code: See coding examples in PDF

Output: [See example output](#)

Check if a number is prime.

Answer: Loop through numbers from 2 to $\sqrt{n}$ and check divisibility.

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 8 – Real-Life & Logical Coding Challenges

Print pattern: * * * * *

Answer: Use nested loops to print each row with increasing stars.

Example code: See coding examples in PDF

Output: [See example output](#)

Celsius to Fahrenheit conversion

Answer: Use formula $F = C \times 9/5 + 32$.

Example code: See coding examples in PDF

Output: [See example output](#)

SECTION 9 – Logical Output-Based Questions

What will `print([])` evaluate to?

Answer: It will evaluate to False because empty sequences are considered False.

Example code: See coding examples in PDF

Output: See example output

Difference between 'is' and '=='?

Answer: '==' checks value equality. 'is' checks object identity (memory location).

Example code: See coding examples in PDF

Output: See example output

SECTION 10 – Practical Short Questions

Difference between local and global variable?

Answer: Local variables are defined inside functions and accessible only there. Global variables are defined outside functions and accessible anywhere.

Example code: See coding examples in PDF

Output: See example output

What is slicing?

Answer: Slicing is accessing a subset of elements from a sequence using [start:end:step].

Example code: See coding examples in PDF

Output: See example output

SECTION 11 – Quick Practice Programs

Print Fibonacci series up to n terms

Answer: Use a loop or recursion to generate series.

Example code: See coding examples in PDF

Output: [See example output](#)

Check if a string is palindrome

Answer: Compare the string with its reverse.

Example code: See coding examples in PDF

Output: [See example output](#)

■ Quick Revision Sheet – Top 25 Questions

1. Difference between list and tuple
2. What is lambda function?
3. Explain decorators
4. What are iterators and generators
5. Difference between is and ==
6. What is `__init__` method?
7. Explain list comprehension
8. What is recursion?
9. What is the use of self?
10. Explain inheritance
11. What are *args and **kwargs?
12. Difference between shallow copy and deep copy
13. What is a package in Python?
14. What is PEP 8?
15. What is the difference between local and global variables?
16. How to open a file in Python?
17. Difference between mutable and immutable data types
18. How to remove duplicates from a list?
19. How to check if a string is palindrome?
20. How to reverse a string?
21. How to find largest element in a list?
22. How to generate Fibonacci series?
23. How to handle exceptions?
24. How to merge two dictionaries?
25. How to check prime numbers?